

SVKM's NMIMS University
Mukesh Patel School of Technology Management and Engineering

Program: B Tech /MBA Tech (Computer Engineering) B Tech (CSBS, Computer Science)/ BTI Computer				Semester: V/ VI /VII/X	
Course: Machine Learning				Code: 702CO1E001	
Teaching Scheme				Evaluation Scheme	
Lecture Hours per week	Practical Hours per week	Tutorial Hours per week	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks- 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Prerequisite - Probability and Statistics, Python Programming					
Course Objective This course provides a concise introduction to the fundamental concepts in machine learning from a practical perspective. Also, it covers the different learning algorithms and paradigms used in Machine Learning.					
Course Outcomes After successful completion of this course, students will be able to - 1. Identify machine learning techniques suitable for a given problem, 2. Solve the problems using various machine learning techniques, 3. Develop an application using machine learning techniques, 4. Evaluate and interpret the results of the algorithms.					
Detailed Syllabus					
Unit	Description				Duration
1.	Machine Learning Fundamentals Terminology, Supervised and Unsupervised Learning with examples, Underfitting / Overfitting, Bias-Variance Trade-off, Model Selection, Applications				02
2.	Exploratory data Analysis Missing Value Treatment, Handling Categorical data: Mapping ordinal features, Encoding class labels, Performing one-hot encoding on nominal features, Outlier Detection and Treatment. Feature Engineering: Variable Transformation and Variable Creation, Selecting meaningful features.				02
3.	Regression Linear regression using Least Squares (analytical approach), Linear regression using Gradient Descent (iterative approach), Multiple linear regression, Polynomial regression				06
4.	Classification Performance Evaluation, Confusion Matrix, Accuracy, Precision, Recall, F1-score, ROC Curves, AUC, k-fold Cross-Validation. Logistic Regression, Naive Bayes Classifier, Support Vector Machines, Neural Networks: Perceptron, Multi-layer Perceptron, Training using Back-propagation,				08



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	Applications	
5.	Tree-Based Methods Basics of Decision Trees, Regression Trees, Classification Trees, Trees v/s Linear Models, Advantages and Disadvantages of Trees, Ensemble techniques- Bagging, Boosting, Random forest	06
6.	Unsupervised Learning Challenges of Unsupervised Learning, Partitioning Methods: K-Means clustering, Dimension Reduction Methods, Principal Component Analysis (PCA), Hierarchical Clustering, Introduction to Recommender systems (Non-personalized and Content-based)	06
	Total	30

Text Books:

1. Judith Hurwitz, Daniel Kirsch, "Machine Learning for dummies", IBM Limited Edition, John Wiley & Sons, Inc., 2018.
2. Trevor Hastie, Robert Tibshirani, Jerome Friedman, "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", 2nd Edition, Springer, 2017.

Reference books

1. Aurelien Geron, "Hands-on Machine Learning with Scikit Learn, Keras and Tensorflow", 2nd Edition, Oreilly Publication, 2019.
2. James, Witten, Hastie, Tibshirani, "Introduction to Statistical Learning", 7th Edition, Springer, 2017.

Laboratory Work

8 to 10 experiments (and a practicum where applicable) based on the syllabus



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